EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

AMINE

1. Among the following which one is not formed in Hoffmann degradation

(1) RNCO (2) $R-\ddot{N}H_2$

(3) $RCO\ddot{N}HBr$ (4) RNC

AM0001

2. $CH_3CH_2CONH_2 \xrightarrow[Br_2]{NaOH} A$,

Aqueous solution of A

(1) Turns blue litmus to red

(2) Turns red litmus to blue

(3) Does not affect the litmus

(4) Decolourise the litmus

AM0002

3. Ethanamine can be obtained if the following compound is heated with [KOH + Br_2]

(1) Ethanamide (2) Methanamide

(3) Propionamide (4) All the above

AM0003

4. $CH_3CONH_2 \xrightarrow[I]{P_2O_5} A \xrightarrow[II]{Na/EtOH} B$

Reaction II is called

(1) Clemensen

(2) Stephen

(3) Mendius

(4) Bauveault–blank reduction

CA0004

5. Tertiary amine is obtained in the reaction :-

(1) Aniline $\xrightarrow{CH_3I} \xrightarrow{CH_3I}$

(2) Aniline $\xrightarrow{CH_3I}$

(3) Nitrobenzene $\xrightarrow{Sn/HCl}$

(4) None of the above

AM0007

6. $C_2H_5NH_2$ cannot be prepared by the reduction of

(1) $C_2H_5NO_2$ (2) $CH_3CH=NOH$

(3) C_2H_5NC (4) CH_3CN

AM0008

7. Gabriel reaction for the synthesis of amines, involves the use of

(1) 1° amide (2) 2° amide

(3) Imides (4) Aliphatic amide

AM0009

8. Gabriel phthalimide reaction is used in the synthesis of

(1) Primary aromatic amines

(2) Secondary amines

(3) Primary aliphatic amines

(4) Tertiary amines

AM0010

9. The reaction : [$C_2H_5Br + NH_3$] is in fact an example of

(1) Ammonolysis only

(2) Nucleophilic substitution only

(3) Ammonolysis as well as nucleophilic substitution

(4) None

HD0011

10. Melting points are normally the highest for

(1) Tertiary amides

(2) Secondary amides

(3) Primary amides

(4) Amines

GC0012

11. Solubility of ethylamine in water is due to

(1) Low molecular weight

(2) Ethyl group is present in ethyl alcohol

(3) Formation of H–bonding with water

(4) Being a derivative of ammonia

GC0013

12. Which of the following compound liberates CO_2 when treated with $NaHCO_3$

(1) $CH_3COCH_2NH_2$

(2) CH_3NH_2

(3) $(CH_3)_4\overset{\oplus}{N}\overset{\ominus}{O}H$

(4) $CH_3\overset{\oplus}{N}H_3\overset{\ominus}{Cl}$

PO0014

13. The product obtained by the alkaline hydrolysis of $C_2H_5—N=C=O$ when treated with t–butyl magnesiumbromide, the compound obtained will be

(1) t–butylamine (2) n–butylamine

(3) Isobutane (4) n–butane

AM0015

14. $C_2H_5NH_2 \xrightarrow[C_6H_5CHO]{HNO_2} a$, $\xrightarrow{} b$, $\xrightarrow{NOCl} c$, $\xrightarrow{C_6H_5SO_2Cl} d$

Which product is a Schiff's base :-

(1) a (2) b (3) c (4) d

AM0016

15. Acidic nature of amino group is shown by the reaction :- @neetquestionpaper

(1) $R–NH_2 + NOCl \rightarrow RCl + N_2 + H_2O$

(2) $2RNH_2 + 2Na \rightarrow 2RNH.Na + H_2$

(3) $R.CH_2NH_2 + HNO_2 \rightarrow R.CH_2OH + N_2 + H_2O$

(4) $R.NH_2 + HCl \rightarrow \overset{\oplus}{R}\overset{}{N}H_3\ \overset{\ominus}{Cl}$

AM0017

16. The reagent used in the conversion of $C_2H_5NH_2$ to C_2H_5Cl would be

(1) SO_2Cl_2 (2) $SOCl_2$

(3) NOCl (4) All

AM0018

17. Hydrogen attached to nitrogen is released in the reaction

(1) $RCONH_2 + NaNH_2$

(2) $RNH_2 + Na$

(3) Both the above

(4) None of the abvoe

GC0019

18. If primary amines are treated with ketones the product is

(1) Urea (2) Guanidine

(3) Amide (4) Schiff's base

CC0020

19. Reactants of reaction – I are

CH_3CONH_2, KOH, Br_2

Reactants of reaction–II are

CH_3NH_2, $CHCl_3$, KOH

The intermediate species of reaction–I and reaction–II are respectively

(1) Carbonium ion, carbene

(2) Nitrene, carbene

(3) Carbene, nitrene

(4) Carbocation, carbanion

AM0021

20. This compound does not respond to carbylamine reaction :-

(1) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–NH_2$

(2) $C_2H_5–NH–C_2H_5$

(3) $CH_3–\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}–NH_2$

(4) $CH_3–\underset{\displaystyle NH_2}{\underset{|}{CH}}–CH_2–CH_3$

AM0023

21. Blue litmus can be turned to red by the compound

(1) ROH (2) RNH_2

(3) $R\overset{\oplus}{N}H_3\ \overset{\ominus}{O}H$ (4) $R\overset{\oplus}{N}H_3\ \overset{\ominus}{Cl}$

AM0024

22. Which one of the following amine compound gives alcohol with HNO_2 :-

(1) N,N–Dimethylaniline (2) Benzylamine

(3) N–methylaniline (4) Aniline

AM0026

ANILINE

23. $C_6H_5NH_2 \xrightarrow{Br_2/CCl_4} ?$ The product is :-

(1) Only o– bromoaniline

(2) 2, 4, 6–triboromoaniline

(3) o–and p–bromoaniline

(4) Only p–bromoaniline

AM0028

24. Aniline is purified by :-

(1) Azeotropic distillation

(2) Steam distillation

(3) distillation in presence of magnesium

(4) Fractional distillation

GC0030

25. Reaction $C_6H_5NH_2 + HAuCl_4 \longrightarrow$

$[C_6H_5\overset{\oplus}{N}H_3]Au\overset{\ominus}{Cl}_4$ shows ... behaviour of aniline :-

(1) Acidic (2) Neutral

(3) Basic (4) Amphoteric

GC0031

26. Aniline on treatment with bromine water yields white precipitate of :-

(1) o–Bromoaniline

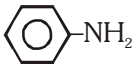
(2) p–Bromonailine

(3) 2, 4, 6–Tribromoaniline

(4) m–Bromoaniline

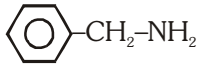
AH0032

27. Which compound does not show diazo reaction:-

(1) 

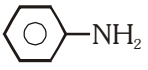
(2) 

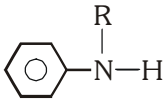
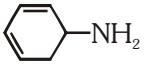
(3) 

(4) 

AM0033

28. Which of the following amines give N–nitroso derivative with $NaNO_2$ and HCl :-

(1) $C_2H_5NH_2$ (2) 

(3)  (4) 

AM0034

29. Which of the following does not reduce Tollen's reagent :-

(1) CH_3CHO (2) HCOOH

(3) C_6H_5NHOH (4) $C_6H_5NH_2$

PO0036

30. Aniline can be obtained by :-

(1) Benzoyl chloride and ammonia

(2) Reduction of benzamide

(3) Phenol and ammonia in presence of $ZnCl_2$

(4) Benzoic anhydride and ammonia

AM0037

31. Aniline on direct nitration produces :-

(1) o–Nitroaniline

(2) m–Nitroaniline

(3) p–Nitroaniline

(4) All

AH0038

32. Nitration of acetanilide followed by hydrolysis gives

(1) o–Nitroaniline only

(2) p–Nitroaniline only

(3) o– & p–Nitroaniline

(4) o–Nitroanilinium ion

AH0039

33. $C_6H_5NH_2 \xrightarrow[0-5°C]{NaNO_2/HCl} A$, Which is the incorrect structure of the product 'A' :-

(1) $[C_6H_5-N=\overset{\oplus}{N}]\overset{\ominus}{Cl}$

(2) $[C_6H_5\overset{\oplus}{N}_2]\overset{\ominus}{Cl}$

(3) $[C_6H_5-\overset{\oplus}{N}\equiv N]\overset{\ominus}{Cl}$

(4) $[C_6H_5-N\equiv\overset{\oplus}{N}]\overset{\ominus}{Cl}$

AM0040

34. Chloroform and ethanolic KOH is used as a reagent in the following reaction :-

(a) Hoffmann carbylamine reaction

(b) Hoffmann degradation reaction

(c) Reimer–Tiemann reaction

(d) Hoffmann mustard oil reaction

Code is :-

(1) Only for a (2) Only for a and b

(3) Only for b and d (4) Only for a and c

AM0041

35. Acetanilide when treated with bromine in acetic acid mainly gives :-

(1) o–Bromoacetanilide

(2) N–Bromoacetanilide

(3) p–Bromoacetanilide

(4) m–Bromoacetanilide

AH0042

36. Aromatic nitriles (ArCN) are not prepared by reaction

(1) ArX + KCN

(2) ArN_2^+ + CuCN

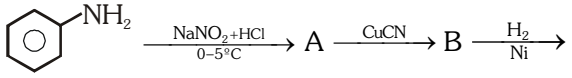
(3) $ArCONH_2 + P_2O_5$

(4) $ArCONH_2 + SOCl_2$

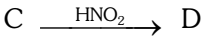
AM0043

NITRO GROUP, CYANIDE & ISOCYANIDE

37. Aniline in a set of reactions yielded end product D



$$\xrightarrow[0-5°C]{NaNO_2+HCl} A \xrightarrow{CuCN} B \xrightarrow[Ni]{H_2}$$



$$C \xrightarrow{HNO_2} D$$

The structure of the product D would be

(1) $C_6H_5CH_2OH$

(2) $C_6H_5CH_2NH_2$

(3) C_6H_5NHOH

(4) $C_6H_5NHCH_2CH_3$

AM0044

38. $\phi—X \xrightarrow{NaNO_2/HCl} C_6H_5N_2Cl \xrightarrow[\Delta]{Water} \phi—Y$,

In the above sequence X and Y are :-

(1) o–, p– and m–directing

(2) o–, p– and o–, p–directing

(3) m and m directing

(4) m and o, p directing

AM0046

39. Which of the following compound gives an explosive on decarboxylation :-

(1) 2,4, 6–Trinitrobenzoicacid

(2) 2, 4–Dinitrobenzoicacid

(3) o–Aminobenzoicacid

(4) o–Hydroxybenzoicacid

AM0047

40. The gas leaked from a storage tank of the Union Carbide plant in Bhopal gas tragedy was:-

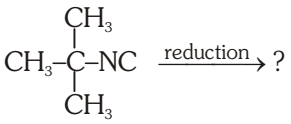
(1) Methylisocyanate

(2) Methylamine

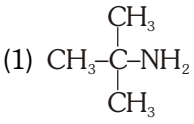
(3) Ammonia

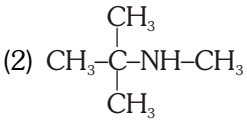
(4) Phosgene

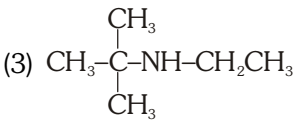
AM0048

41. 

$$CH_3–\overset{CH_3}{\underset{CH_3}{C}}–NC \xrightarrow{reduction} ?$$

(1) 

(2) 

(3) 

(4) None

AM0049

42. Reaction of RCN with sodium and alcohol leads to the formation of :-

(1) $RCONH_2$ (2) $RCOO^-NH_4^+$

(3) RCH_2NH_2 (4) $R(CH_2)_3NH_2$

AM0050

43. $C_6H_5NO_2 \xrightarrow{SnCl_2/HCl} A \xrightarrow[0°C]{NaNO_2/HCl} B$; In the above sequence Benzene from B, is suitably obtained by using :-

(1) Ethanol (2) H_3PO_2

(3) Both the above (4) Methanol

AM0051

44. Which reagent is used to get iodo benzene from benzene diazonium acid sulphate $[C_6H_5N_2HSO_4]$:

(1) CuBr, Δ (2) Cu powder + HI

(3) KI, Δ (4) None

AM0052

45. Which of the following is used as a solvent in the Friedel–Crafts reaction :-

(1) Toluene (2) Nitrobenzene

(3) Benzene (4) Aniline

AH0053

46. In the Sandmeyer's reaction, —N═N—X group of diazonium salt is replaced by :-

(1) Halide group (2) Nitro group

(3) —OH group (4) —$NHNH_2$ group

AM0055

EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**2**	**3**	**3**	**1**	**3**	**3**	**3**	**3**	**3**	**3**	**4**	**3**	**2**	**2**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**3**	**3**	**4**	**2**	**2**	**4**	**2**	**2**	**2**	**3**	**3**	**4**	**3**	**4**	**3**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**4**	**3**	**1**	**4**	**3**	**1**	**1**	**2**	**1**	**1**	**2**	**3**	**3**	**3**	**2**
Que.	46														
Ans.	**1**														

EXERCISE-III (Analytical Questions)

Master Your Understanding

1. $A \xrightarrow{NH_3} B \xrightarrow{\Delta} C \xrightarrow[KOH]{Br_2+} CH_3CH_2NH_2$

A is :-

(1) CH_3COOH

(2) $CH_3CH_2CH_2COOH$

(3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH$

(4) CH_3CH_2COOH

AM0083

2. The correct set of the products obtained in the following reactions

$RCN \xrightarrow{\text{reduction}} (A)$, $RCN \xrightarrow[(ii)H_2O]{(i)CH_3MgBr} (B)$,

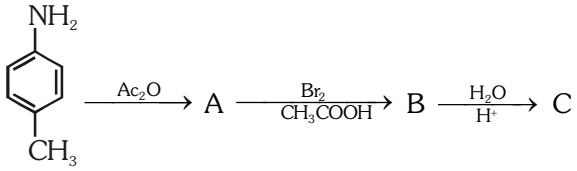
$RNC \xrightarrow{\text{hydrolysis}} (C)$, $RNH_2 \xrightarrow{HNO_2} (D)$

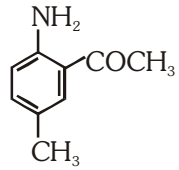
The answer is

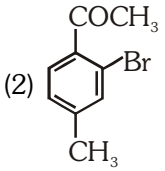
	A	B	C	D
(1)	2°Amine	Methyl ketone	1° Amine	Alcohol
(2)	1°Amine	Methyl ketone	1° Amine	Alcohol
(3)	2°Amine	Methyl ketone	2° Amine	Acid
(4)	1°Amine	Methyl ketone	2° Amine	Aldehyde

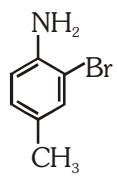
AM0084

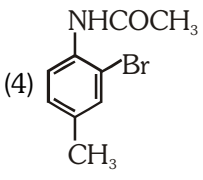
3. The final product C, obtained in this reaction would be



(1) 

(2) 

(3) 

(4) 

AH0085

4. $C_2H_5Br \xrightarrow{AgCN} A \xrightarrow{H_3O^+} HCOOH + B$;

$B \xrightarrow[KOH]{CHCl_3} A \xrightarrow{\text{Reduction}} C$

A, B, C respectively in the above sequence are

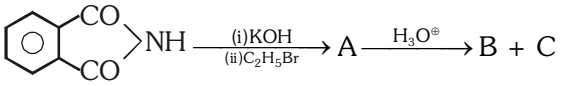
(1) Ethane amine, methane nitrile and diethyl amine

(2) Carbyl amino ethane, ethane amine and primary amine

(3) Ethyl isocyanide, ethyl amine and methyl isocyanate

(4) Carbylamino ethane, ethanamine and ethyl methyl amine

AM0086

5. 

B and C in the above sequence are

(1) Benzoic acid + aniline

(2) Phthalic acid + ethylamine

(3) Phthalic acid + aniline

(4) Benzoic acid + ethylamine

AM0087

6. The end–product in the reaction sequence would be :

Ethyl amine $\xrightarrow{HNO_2} A \xrightarrow{PCl_5} B \xrightarrow{NH_3} C$

(1) Ethyl cyanide

(2) Ethyl amine

(3) Methyl amine

(4) Acetamide

AM0088

7. $A \xrightarrow{H_2NOH} B \xrightarrow{\text{reduction}} C \xrightarrow{NOCl} CH_3CH_2Cl$,

A and C in the above sequence respectively are:-

(1) Methanal, Methylamine

(2) Acetone, ethanamine

(3) Ethanal, dimethylamine

(4) Acetaldehyde, ethylamine

CC0089

8. A compound of mol. wt. 180 gm is acetylated to give a compound of mol. wt. 390. The number of amino groups in the compound are :-

(1) 2 (2) 4

(3) 5 (4) 6

AM0090

9. In the reaction sequence A, B and C are :-

$A \xrightarrow[HCl]{SnCl_2} B \xrightarrow[0°C]{NaNO_2/HCl} C \xrightarrow{H_2O/\Delta} C_6H_5OH$

(1) Benzene, nitrobenzene, aniline

(2) Nitrobenzene, aniline and azo–compound

(3) Nitrobenzene, benzene, aniline

(4) Benzene, amino compound, aniline

AM0091

10. Which one of the following tests can be used to identify primary amino group in a given organic compound

(1) Iodoform test

(2) Victor Meyer's test

(3) Carbylamine reaction

(4) Libermann's reaction

PO0092

11. $CH_3Cl \xrightarrow{KCN} \xrightarrow{+4H} A \xrightarrow{NaNO_2/HCl}$ Ethanol + ?

Apart from ethanol as the main product, the other products would be

(1) $H_2C=CH_2$ (2) CH_3CH_2–ONO

(3) CH_3CH_2–Cl (4) All the above

AM0094

12. Identify B, X and R respectively in the following sequence of reactions

$C_2H_5MgBr \xrightarrow{ClCN} A \xrightarrow{H_3O^+} \boxed{B}$,

$CH_3COCH_3 \xrightarrow[NaOH]{I_2} \boxed{X} \xrightarrow{Ag} CH\equiv CH$

$C_6H_5NH_2 \xrightarrow[0-5°C]{NaNO_2/HCl} P \xrightarrow{CuCN} Q \xrightarrow[EtOH]{Na} \boxed{R}$

(1) C_2H_5COOH, CHI_3, $C_6H_5CH_2NH_2$

(2) C_2H_5COOH, CH_3I, C_6H_5COOH

(3) $C_2H_5CH_2NH_2$, CH_3I, C_6H_5COOH

(4) C_2H_5COOH, C_2H_5I, $C_6H_5CONH_2$

AM0095

13. Using Fe/HCl, which one of the following reaction is possible [Here $\phi = C_6H_5$]

(1) ϕ–$NO_2 \rightarrow \phi$–NH–NH–ϕ

(2) ϕ–$NO_2 \rightarrow \phi$–NH_2

(3) ϕ–$NO_2 \rightarrow \phi$–NHOH

(4) ϕ–$NO_2 \rightarrow \phi$–N=N–ϕ

AM0096

14. Total structural isomeric amines possible for molecular formula $C_4H_{11}N$

(1) 7 (2) 8 (3) 9 (4) 6

NC0109

15. $C_2H_5NH_2 \xrightarrow{NOCl} A \xrightarrow{NaCN} B \xrightarrow{H_2/Ni} C$

what will be "C"

(1) $CH_3 - CH_2 - CH_2 - NH_2$

(2) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - NH_2$

(3) $CH_3\,CH_2 - CH = NH$

(4) $H_5C_2 - \underset{\displaystyle CH_2 - NH_2}{\underset{|}{N}} - N = 0$

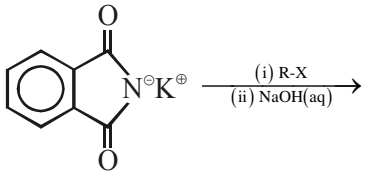
AM0110

16. Which of following will not gives only primary amine

(1) $C_2H_5X \xrightarrow{NH_3(Alc)}$

(2) $CH_3 - CN \xrightarrow[C_2H_5OH]{Na(Hg)}$

(3) $CH_3 - \overset{\displaystyle O}{\overset{\|}{C}} - NH_2 \xrightarrow[(ii)\,H_2O]{(i)\,LiAlH_4}$

(4) 

AM0111

17. Basic nature of amine in aqueous medium can be explained by :-

(1) Inductive effect (2) Solvation effect

(3) Steric hinderance (4) All of these

AM0112

18. Which of following is correct regarding basic nature:-

(1) $R - NH_2 > R_2NH > R_3N$ (Gaseous phase)

(2) $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$ (Aqueous solutions)

(3) $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5$–$NH_2 > NH_3$ (Aqueous solutions)

(4) $(C_2H_5)_2NH > PhNH_2 > Ph - NH - CH_3$ (Aqueous solution)

GC0113

19. What is decreasing order of H-bonding in water for alkyl substituted ammonium cation ?

(1) $R\overset{\oplus}{N}H_3 > R_2\overset{\oplus}{N}H_2 > R_3\overset{\oplus}{N}H$

(2) $R_3\overset{\oplus}{N}H > R_2\overset{\oplus}{N}H_2 > R\overset{\oplus}{N}H_3$

(3) $R_2\overset{\oplus}{N}H_2 > R_3\overset{\oplus}{N}H > R\overset{\oplus}{N}H_3$

(4) $R_2\overset{\oplus}{N}H_2 > R\overset{\oplus}{N}H_3 > R_3\overset{\oplus}{N}H$

AM0114

20. Which of following reaction is not acylation ?

(1) $C_2H_5NH_2 + CH_3-\overset{O}{\overset{\|}{C}}-Cl \longrightarrow$

(2) $C_2H_5NH_2 + CH_3-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-CH_3 \longrightarrow$

(3) $C_2H_5NH_2 + CH_3\overset{O}{\overset{\|}{C}}-OCH_3 \longrightarrow$

(4) $C_2H_5NH_2 + CH_3-\overset{O}{\overset{\|}{C}}-OH \longrightarrow$

AM0115

21. Which of following will not give acylation reaction?

(1) $H_5C_2-NH_2$ (2) $H_5C_2-NH-C_2H_5$

(3) $C_2H_5-\underset{\underset{C_2H_5}{|}}{\overset{C_2H_5}{\overset{|}{N}}}$ (4) $H_5C_6-NH_2$

AM0116

22. Which of following is not correct match?

(1) $H_5C_2NH_2 + CH_3-\overset{O}{\overset{\|}{C}}-Cl \longrightarrow H_5C_2-NH-\overset{O}{\overset{\|}{C}}-CH_3 + HCl$
(Acylation)

(2) $H_5C_2NH_2 + H_5C_6-\overset{O}{\overset{\|}{C}}-Cl \longrightarrow H_5C_2-NH-\overset{O}{\overset{\|}{C}}-C_6H_5 + HCl$
(Benzoylation)

(3) $(H_5C_2)_2NH + CHCl_3 + 3KOH \longrightarrow H_5C_2-NC + 3KCl + 3H_2O$
(Carbyl amine reaction)

(4) $H_5C_2-NH_2 + CH_3COOH \rightarrow H_5C_2\overset{\oplus}{N}H_3\,CH_3COO^-$
(Salt formation or Neutralisation)

AM0117

23. Which of following can be use to distinguish primary, secondary and tertiary amine?

(1) Carbyl amine reaction

(2) Reaction with Nitrous acid

(3) Reaction with Hinsberg's reagent

(4) Both 2 & 3

PO0118

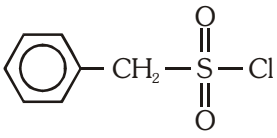
24. Which of following is insoluble in alkali?

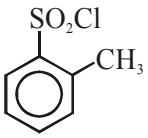
(1) $H_5C_6-\overset{O}{\overset{\|}{\underset{O}{\overset{\|}{S}}}}-N-C_2H_5$ (2) $H_5C_6-\overset{O}{\overset{\|}{\underset{O}{\overset{\|}{S}}}}-N-CH_3$

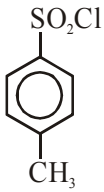
(3) $H_5C_6-\overset{O}{\overset{\|}{\underset{O}{\overset{\|}{S}}}}-N-C_2H_5$ (4) All of these

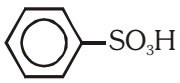
AM0119

25. These days benzene sulphonyl chloride replaced by which compound in Hinsberg test :-

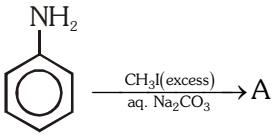
(1) 

(2) 

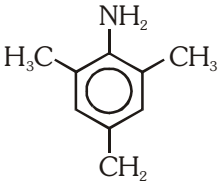
(3) 

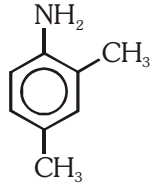
(4) 

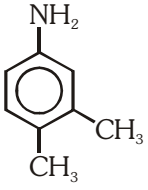
AM0120

26. 

What will be "A"

(1) 

(2) 

(3) 

(4) $Ph-\overset{\oplus}{N}(CH_3)_3\overset{\ominus}{I}$

AM0121

27. Total number of isomers of molecular formula C_3H_9N which will liberate N_2 gas on treatment with nitrous acid?

(1) 2 (2) 3 (3) 4 (4) 1

AM0122

28. Which of following is water insoluble and stable at room temperature.

(1) $C_6H_5N_2^{\oplus}Cl^{\ominus}$ (2) $C_6H_5N_2^{\oplus}BF_4^{\ominus}$

(3) $C_6H_5N_2^{\oplus}HSO_4^{\ominus}$ (4) All

AM0123

29. Which of following reaction do not displace N_2 from benzenediazonium salt?

(1) $ArN_2^{\oplus}Cl^- \xrightarrow{Cu_2Cl_2/HCl}$

(2) $ArN_2^{\oplus}Cl^- \xrightarrow{Cu/HCl}$

(3) $ArN_2^{\oplus}Cl^- + KI \longrightarrow$

(4) $ArN_2^{\oplus}Cl^- + HBF_4 \longrightarrow$

AM0124

30. Which of following group can not be introduced in the benzene ring by sandmeyer reaction?

(1) –Cl (2) –Br (3) –CN (4) –I

AM0125

31. Formation of benzene from $C_6H_5N_2^{\oplus}Cl^-$ by use of C_2H_5OH is example of :-

(1) Redox reaction

(2) Electrophilic substitution reaction

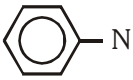
(3) Nucleophilic substitution reaction

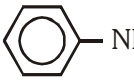
(4) Elimination reaction

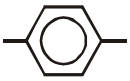
AM0126

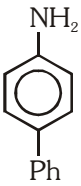
32. $A + HBF_4 \longrightarrow B \xrightarrow[Cu,\ \Delta]{NaNO_2} C \xrightarrow{Fe/HCl} D \xrightarrow[H^{\oplus}]{\Delta} E$

What is "E"

(1)  N = N – Ph

(2)  NH – H = N – Ph

(3) Ph – N = N  NH_2

(4) 

AM0127

33. Diazo coupling reaction can be use to distinguish

(1) Aniline and Benzyl amine

(2) Ethylamine and N-methyl aniline

(3) Aniline and Phenol

(4) All of these

AM0128

34. What is correct sequence of reagent to form 2,4,6-tibromofluorobenzene from Aniline?

(1) $NaNO_2/HCl$ (0-5°C), HBF_4/Δ, Br_2/H_2O

(2) Br_2/H_2O, $NaNO_2/HCl$ (0-5°C), HBF_4/Δ

(3) $NaNO_2/HCl$ (25°C), HBF_4/Δ, Br_2/H_2O

(4) Br_2/H_2O, $NaNO_2/HCl$ (0-5°C), HBF_4 (Room temperature)

AM0129

EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**2**	**3**	**4**	**2**	**2**	**4**	**3**	**2**	**3**	**4**	**1**	**2**	**2**	**1**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**1**	**4**	**3**	**1**	**4**	**3**	**3**	**4**	**1**	**3**	**4**	**1**	**2**	**4**	**4**
Que.	31	32	33	34											
Ans.	**1**	**3**	**4**	**2**											